

Pooled-Panel PCA

Institutional axes that hold up across 1970-2025

Institutional Complexity Project

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Contents

Why pool	2
Data	2
Indicator selection	2
Country-year row construction	3
Imputation and z-scoring	4
Fit the PCA	4
How much variance does each PC explain?	5
What does each PC capture?	5
PC1 — top positive loadings	6
PC1 — top negative loadings	6
PC2 — top positive loadings	6
PC2 — top negative loadings	6
PC3 — top positive loadings	7
PC3 — top negative loadings	7
PC4 — top positive loadings	7
PC4 — top negative loadings	8
PC2 in detail: top and bottom 15 loadings with codebook definitions	8
PC2 — top 15 positive loadings	8
PC2 — top 15 negative loadings	11
Loading mass by V-Dem category, per PC	14
Sample country trajectories	15

Two-axis snapshot at year endpoints	17
Key takeaways: PC1 and PC2 as analytical tools	18
PC1: openness of the regime (the democracy axis)	18
PC2: stateness — how much of social life the state encompasses	18
How to deploy PC1 and PC2 in case studies	19
Persist outputs	19
Notes and what to read into this	19

Why pool

The country-space writeup builds a fresh PCA on **only** the 2025 V-Dem cross-section. The cambodia-analysis trajectory chart then projects 1970-2024 data through that 2025 PCA and reads off PC2 over time. The driver decomposition on Cambodia 2006-2025 surfaced the cost of that choice: 9 of the 12 top PC2 movers belong to V-Dem’s Digital Society Survey (v2sm*), a module that didn’t exist before 2000 and accounts for ~39% of the absolute contribution mass to PC2 (vs. its 18% share of the feature set). Fitting on 2025 means the PCA’s axes inherit 2025’s variance structure — a structure dominated by a post-2000 module that for two-thirds of the trajectory we project through is entirely imputed.

A pooled-panel PCA addresses this directly. We stack every (country, year) observation from 1970-2025 into one tall matrix, z-score across the pool, and fit one PCA. The loadings then reflect variance over the **whole window we care about**, not the present. Indicators with sparse historical coverage contribute less to the loading geometry because their pool-wide variance is small (they’re approximately constant at the imputed mean for half the panel).

What we lose: the exact scaling of axis units in any single year is no longer a “within-year SD” — it’s a “within-pool SD”. That’s fine for the questions we want to ask of these axes (“how big a move is Cambodia 2006 → 2025 relative to the kinds of moves countries make in this dataset?”).

Data

Panel window: 1970-2025

Indicator selection

We mirror the country-space writeup’s filter:

- vartype "C" only (component-type IRT scores; aggregates would double-count)
- drop the two discrete-categorical components V-Dem ships (v2regopploc, v2regsuploc)
- drop the 18 conceptually-redundant indicators flagged in outputs/redundancy_clusters_2025.csv
- drop the suffix variants (_code_low, _code_high, _sd, _osp, _ord, _mean, _nr, _rec) and per-rater columns (_1, _2, ...)

The new piece: a **panel-wide coverage filter**. An indicator must be observed in at least 50% of (country, year) cells across the 1970-2025 panel. This is the rule that decides whether the Digital Society Survey indicators (v2sm*, coded only 2000-2025) survive — they’re observed in roughly 46% of years, which is right at the threshold.

Candidate indicators: 251

Pass $\geq 50\%$ pool coverage: 190

Table 1: How the panel-coverage filter cuts each V-Dem category.

category	Dropped (<50% pool cov)	Kept	Total
Digital Society / Social Media	32	0	32
Elections	8	22	30
Legislature	17	10	27
Political Equality	0	22	22
Civil Liberties	1	20	21
Executive	0	20	20
Education	0	19	19
Media	0	15	15
Mobilization & Academic	0	14	14
Political Parties	0	11	11
Judiciary	0	9	9
Regime	0	7	7
State Capacity	2	5	7
Civil Society	0	6	6
Deliberation	0	6	6
Sovereignty	0	3	3
Other	1	0	1
Technology & Media Environment	0	1	1

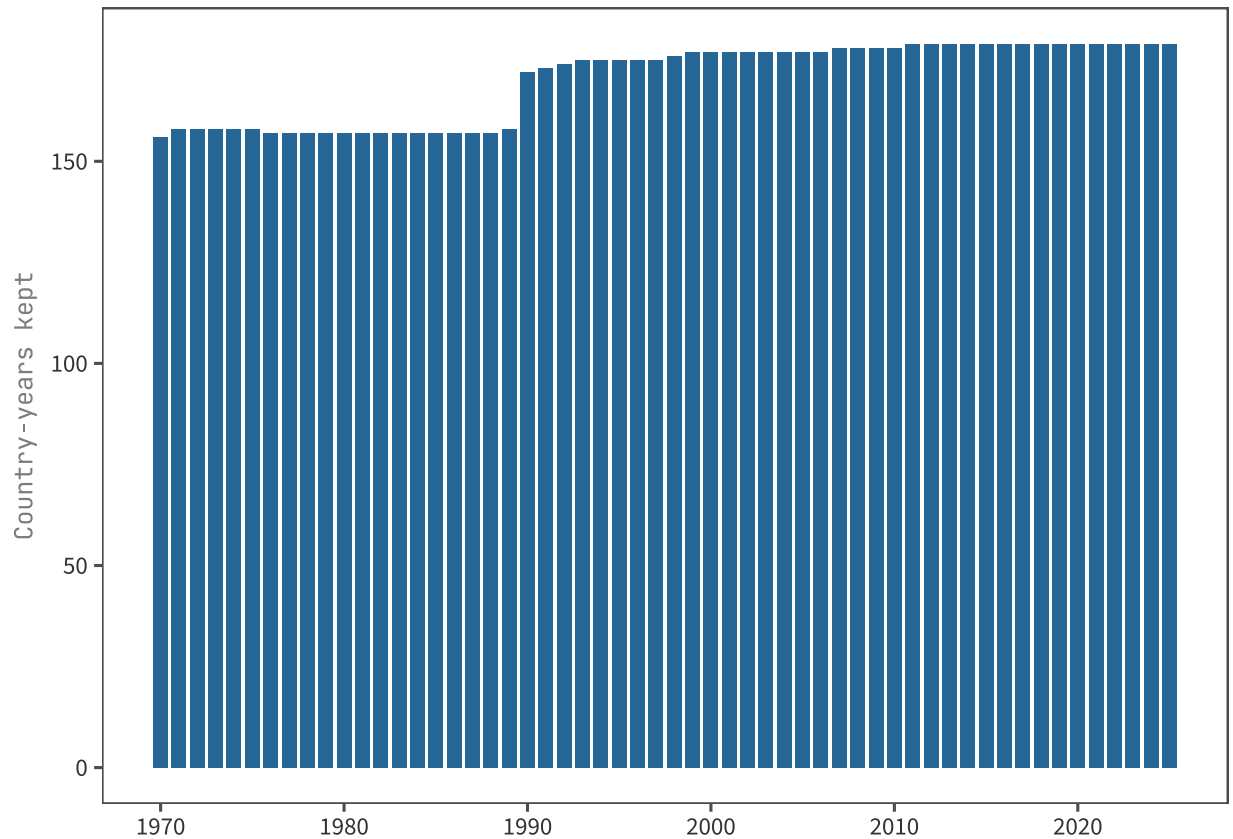
v2sm* (Digital Society) candidates: 32

v2sm* surviving the 50% filter: 0

The Digital Society block is largely culled; whatever survives is whatever V-Dem coded retroactively for some pre-2000 years.

Country-year row construction

Country-year rows kept: 9528 (of 9528 in window)



Imputation and z-scoring

Mean-impute remaining NAs at the **pool** level (not within-year), then z-score each indicator across the pool. The pool-mean fill is the most defensible neutral value for a country-year that's missing a feature: it treats the imputed cell as "an average country-year on this dimension."

Final matrix: 9528 country-years x 190 features

Mean per-feature SD after z-scoring (sanity, should be 1): 1.000

Fit the PCA

How much variance does each PC explain?

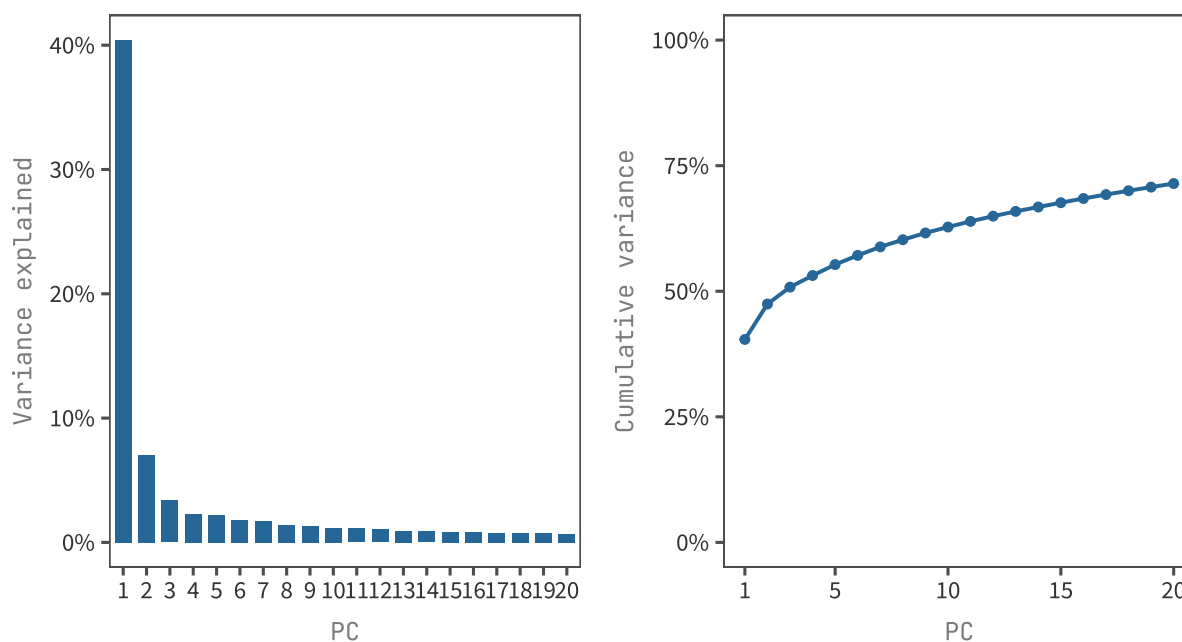


Table 2: First 10 principal components.

PC	Eigenvalue (var)	Variance explained	Cumulative variance
1	76.78	40.4%	40.4%
2	13.40	7.1%	47.5%
3	6.39	3.4%	50.8%
4	4.39	2.3%	53.1%
5	4.14	2.2%	55.3%
6	3.47	1.8%	57.1%
7	3.23	1.7%	58.8%
8	2.71	1.4%	60.3%
9	2.53	1.3%	61.6%
10	2.26	1.2%	62.8%

The first PC absorbs the dominant democracy/autocracy axis and is large; the next several PCs each capture meaningful but smaller chunks. We carry PC1-PC4 forward.

What does each PC capture?

For each of the first four PCs we list the indicators with the largest positive and negative loadings, grouped by V-Dem category. A loading is the indicator's coefficient in that PC's direction in the standardized feature space — ie how much a one-SD move in that indicator shifts a country-year along this PC.

PC1 — top positive loadings

Code	Indicator	Category	Loading
v2mecenefm	Government censorship effort — Media	Media	0.103
v2cldiscm	Freedom of discussion for men	Civil Liberties	0.102
v2cltort	Freedom from torture	Civil Liberties	0.102
v2caassemb	Freedom of peaceful assembly	Mobilization & Academic	0.102
v2elembaut	EMB autonomy	Elections	0.102
v2csreprss	CSO repression	Civil Society	0.102
v2meharjrn	Harassment of journalists	Media	0.102
v2cltrnslw	Transparent laws with predictable enforcement	Civil Liberties	0.102

PC1 — top negative loadings

Code	Indicator	Category	Loading
v2medentrain	Control of entertainment content	Media	-0.082
v2edtefire	Political teacher firing	Education	-0.078
v2medstateprint	State-owned print media	Media	-0.077
v2exl_legitlead	Person of the Leader	Executive	-0.077
v2exdfvths	HOS veto power in practice	Executive	-0.076
v2medpolstate	Political influence, state-owned media	Media	-0.074
v2castate	Engagement in state-administered mass organizations	Mobilization & Academic	-0.071
v2caautmob	Mobilization for autocracy	Mobilization & Academic	-0.071

PC2 — top positive loadings

Code	Indicator	Category	Loading
v2peedueq	Educational equality	Political Equality	0.160
v2pehealth	Health equality	Political Equality	0.152
v2elvotbuy	Election vote buying	Elections	0.147
v2elsnlsff	Subnational election unevenness	Elections	0.145
v2peapsecon	Access to public services distributed by socio-economic position	Political Equality	0.145
v2lgcrpt	Legislature corrupt activities	Legislature	0.138
v2peapsgeo	Access to public services distributed by urban-rural location	Political Equality	0.136
v2elpeace	Election other electoral violence	Elections	0.130

PC2 — top negative loadings

Code	Indicator	Category	Loading
v2cademmob	Mobilization for democracy	Mobilization & Academic	-0.152
v2clstown	State ownership of economy	Civil Liberties	-0.142
v2caviol	Political violence	Mobilization & Academic	-0.137
v2psparban	Party ban	Political Parties	-0.123
v2csantimv	CSO anti-system movements	Civil Society	-0.123
v2regoppgroupsize	Regime opposition groups size	Regime	-0.112
v2csptrcpt	CSO participatory environment	Civil Society	-0.112
v2mebias	Media bias	Media	-0.110

PC3 — top positive loadings

Code	Indicator	Category	Loading
v2clgencl	Gender equality in respect for civil liberties	Civil Liberties	0.172
v2mefemjrn	Female journalists	Media	0.163
v2cagenmob	Mass mobilization	Mobilization & Academic	0.157
v2cademmob	Mobilization for democracy	Mobilization & Academic	0.150
v2psprbrch	Party branches	Political Parties	0.146
v2edpatriot	Patriotic education in the curriculum	Education	0.141
v2exl_legitperf	Performance legitimization	Executive	0.135
v2edscpatriotcb	Celebration of patriotic symbols	Education	0.132

PC3 — top negative loadings

Code	Indicator	Category	Loading
v2exdjcbhg	HOG appoints cabinet in practice	Executive	-0.154
v2exdfdshg	HOG dismisses ministers in practice	Executive	-0.153
v2exjdshg	HOG dissolution in practice	Executive	-0.139
v2jucorrdc	Judicial corruption decision	Judiciary	-0.131
v2excrtps	Public sector corrupt exchanges	Executive	-0.128
v2lgcrrpt	Legislature corrupt activities	Legislature	-0.124
v2exbribe	Executive bribery and corrupt exchanges	Executive	-0.123
v2exthftps	Public sector theft	Executive	-0.120

PC4 — top positive loadings

Code	Indicator	Category	Loading
v2svdomaut	Domestic autonomy	Sovereignty	0.189
v2svinlaut	International autonomy	Sovereignty	0.160

Code	Indicator	Category	Loading
v2exdfcbhs	HOS appoints cabinet in practice	Executive	0.155
v2exl_legitperf	Performance legitimization	Executive	0.149
v2cldmovem	Freedom of domestic movement for men	Civil Liberties	0.144
v2edscpatriot	Presence of patriotic symbols in schools	Education	0.141
v2exl_legitlead	Person of the Leader	Executive	0.128
v2exdfvths	HOS veto power in practice	Executive	0.124

PC4 — top negative loadings

Code	Indicator	Category	Loading
v2regproreg	Strongest pro-regime preferences	Regime	-0.244
v2regimpgroup	Regime most important support group	Regime	-0.238
v2regpower	Most powerful group in affecting regime duration and change	Regime	-0.223
v2cacamps	Political polarization	Mobilization & Academic	-0.175
v2psprbrch	Party branches	Political Parties	-0.171
v2caviol	Political violence	Mobilization & Academic	-0.160
v2regimpoppgroup	Regime most important opposition group	Regime	-0.158
v2pscohesv	Legislative party cohesion	Political Parties	-0.158

PC2 in detail: top and bottom 15 loadings with codebook definitions

PC2 is the second-most-important institutional axis after democracy and the one whose interpretation requires care — several top contributors have inverted scales, so the loading sign and the meaning of “high indicator score” don’t line up the way the indicator names suggest. We lay out the top and bottom 15 PC2 loadings with V-Dem’s full question text and response options so the reading developed in the takeaways section below is independently verifiable from primary materials.

PC2 — top 15 positive loadings

These pull a country-year toward the positive pole of PC2.

v2peedueq — Educational equality (loading +0.160)

Question. To what extent is high quality basic education guaranteed to all, sufficient to enable them to exercise their basic rights as adult citizens?

Responses. 0: Extreme. Provision of high quality basic education is extremely unequal and at least 75 percent (%) of children receive such low-quality education that undermines their ability to exercise their basic rights as adult citizens. 1: Unequal. Provision of high quality basic education is extremely unequal and at least 25 percent (%) of children receive such low-quality education that undermines their ability to exercise their basic rights as adult citizens. 2: Somewhat equal. Basic education is relatively equal in quality but ten to 25 percent (%) of children receive such low-quality education that undermines their ability to exercise their basic rights as adult citizens. 3:

Relatively equal. Basic education is overall equal in quality but five to ten percent (%) of children receive such low-quality education that probably undermines their ability to exercise their basic rights as adult citizens. 4: Equal. Basic education is equal in quality and less than five percent (%) of children receive such low-quality education that probably undermines their ability to exercise their basic rights as adult citizens.

v2pehealth — Health equality (loading +0.152)

Question. To what extent is high quality basic healthcare guaranteed to all, sufficient to enable them to exercise their basic political rights as adult citizens?

Responses. 0: Extreme. Because of poor-quality healthcare, at least 75 percent (%) of citizens' ability to exercise their political rights as adult citizens is undermined. 1: Unequal. Because of poor-quality healthcare, at least 25 percent (%) of citizens' ability to exercise their political rights as adult citizens is undermined. 2: Somewhat equal. Because of poor-quality healthcare, ten to 25 percent (%) of citizens' ability to exercise their political rights as adult citizens is undermined. 3: Relatively equal. Basic health care is overall equal in quality but because of poor-quality healthcare, five to ten percent (%) of citizens' ability to exercise their political rights as adult citizens is undermined. 4: Equal. Basic health care is equal in quality and less than five percent (%) of citizens cannot exercise their basic political rights as adult citizens.

v2elvotbuy — Election vote buying (loading +0.147)

Question. In this national election, was there evidence of vote and/or turnout buying?

Responses. 0: Yes. There was systematic, widespread, and almost nationwide vote/turnout buying by almost all parties and candidates. 1: Yes, some. There were non-systematic but rather common vote-buying efforts, even if only in some parts of the country or by one or a few parties. 2: Restricted. Money and/or personal gifts were distributed by parties or candidates but these offerings were more about meeting an 'entry-ticket' expectation and less about actual vote choice or turnout, even if a smaller number of individuals may also be persuaded. 3: Almost none. There was limited use of money and personal gifts, or these attempts were limited to a few small areas of the country. In all, they probably affected less than a few percent of voters. 4: None. There was no evidence of vote/turnout buying.

v2elsnlsff — Subnational election unevenness (loading +0.145)

Question. Does the freeness and fairness of subnational elections vary across different areas of the country?

Responses. 0: Yes. Subnational elections in some areas of the country are significantly more free and fair (or, alternatively, significantly less free and fair) than subnational elections in other areas of the country. 1: Somewhat. Subnational elections in some areas of the country are somewhat more free and fair (or, alternatively, somewhat less free and fair) than subnational elections in other areas of the country. 2: No. Subnational elections in most or all areas of the country are equally free and fair (or, alternatively, equally not free and not fair).

v2peapsecon — Access to public services distributed by socio-economic position (loading +0.145)

Question. Is access to basic public services, such as order and security, primary education, clean water, and health-care, distributed equally according to socioeconomic position?

Responses. 0: Extreme. Because of poverty or low income, 75 percent (%) or more of the population lack access to basic public services of good quality. 1: Unequal. Because of poverty or low income, 25 percent (%) or more of the population lack access to basic public services of good quality. 2: Somewhat Equal. Because of poverty or low income, 10 to 25 percent (%) of the population lack access to basic public services of good quality. 3: Relatively Equal. Because of poverty or low income, 5 to 10 percent (%) of the population lack access to basic public services of good quality. 4: Equal. Because of poverty or low income, less than 5 percent (%) of the population lack access to basic public services of good quality.

v2lgcrpt — Legislature corrupt activities (loading +0.138)

Question. Do members of the legislature abuse their position for financial gain?

Responses. 0: Commonly. Most legislators probably engage in these activities. 1: Often. Many legislators probably engage in these activities. 2: Sometimes. Some legislators probably engage in these activities. 3: Very occasionally. There may be a few legislators who engage in these activities but the vast majority do not. 4: Never, or hardly ever.

v2peapsgeo — Access to public services distributed by urban-rural location (loading +0.136)

Question. Is access to basic public services, such as order and security, primary education, clean water, and health-care, distributed equally across urban and rural areas?

Responses. 0: Extreme. Because they live in rural areas, 75 percent (%) or more of the population lack access to basic public services of good quality. 1: Unequal. Because they live in rural areas, 25 percent (%) or more of the population lack access to basic public services of good quality. 2: Somewhat Equal. Because they live in rural areas, 10 to 25 percent (%) of the population lack access to basic public services of good quality. 3: Relatively Equal. Because they live in rural areas, only 5 to 10 percent (%) of the population lack access to basic public services of good quality. 4: Equal. Because they live in rural areas, less than 5 percent (%) of the population lack access to basic public services of good quality. 5: Rural-Bias: Because they live in urban areas, 25% or more of the population lack access to basic public services of good quality.

v2e1peace — Election other electoral violence (loading +0.130)

Question. In this national election, was the campaign period, election day, and post-election process free from other types not by the government, the ruling party, or their agents) of violence related to the conduct of the election and the campaigns (but not conducted by the government and its agents)?

Responses. 0: No. There was widespread violence between civilians occurring throughout the election period, or in an intense period of more than a week and in large swaths of the country. It resulted in a large number of deaths or displaced refugees. 1: Not really. There were significant levels of violence but not throughout the election period or beyond limited parts of the country. A few people may have died as a result, and some people may have been forced to move temporarily. 2: Somewhat. There were some outbursts of limited violence for a day or two, and only in a small part of the country. The number of injured and otherwise affected was relatively small. 3: Almost. There were only a few instances of isolated violent acts, involving only a few people; no one died and very few were injured. 4: Peaceful. No election-related violence between civilians occurred.

v2medstatebroad — State-owned broadcast media (loading +0.126)

Question. Out of the top four national broadcast media with the largest audience, how many are state-owned?

Responses. 0: There are no state-owned broadcast media outlets. 1: State-owned outlets make up a minority of media broadcast outlets. 2: There is an equal share of state- and non-state owned broadcast media outlets. 3: State-owned outlets make up the majority of broadcast media outlets. 4: All broadcast media outlets are state-owned.

v2c1rgunev — Subnational civil liberties unevenness (loading +0.125)

Question. Does government respect for civil liberties vary across different areas of the country?

Responses. 0: Yes. Government officials in some areas of the country respect civil liberties significantly more (or, alternatively, significantly less) than government officials in other areas of the country. 1: Somewhat. Government officials in some areas of the country respect civil liberties somewhat more (or, alternatively, somewhat less) than government officials in other areas of the country. 2: No. Government officials in most or all areas of the country equally respect (or, alternatively, equally do not respect) civil liberties.

v2peasjsoecon — Access to state jobs by socio-economic position (loading +0.124)

Question. Are state jobs equally open to qualified individuals regardless of socio-economic position?

Responses. 0: Extreme. Because of poverty or low income, 75 percent (%) or more of the population, even if qualified, lack access to state jobs. 1: Unequal. Because of poverty or low income, makes 25 percent (%) or more of the population, even if qualified, lack access to state jobs. 2: Somewhat Equal. Because of poverty or low income, 10 to 25 percent (%) of the population, even if qualified, lack access to state jobs. 3: Relatively Equal. Because of poverty or low income, 5 to 10 percent (%) of the population, even if qualified, lack access to state jobs. 4: Equal. Because of poverty or low income, less than 5 percent (%) of the population, even if qualified, lack access to state jobs.

v2peapssoc — Access to public services distributed by social group (loading +0.124)

Question. Are basic public services, such as order and security, primary education, clean water, and healthcare, distributed equally across social groups?

Responses. 0: Extreme. Because of their social group, 75 percent (%) or more of the population lack access to basic public services of good quality. 1: Unequal. Because of their social group, 25 percent (%) or more of the population lack access to basic public services of good quality. 2: Somewhat Equal. Because of their social group, 10 to 25 percent (%) of the population lack access to basic public services of good quality. 3: Relatively Equal. Because of their social group, only 5 to 10 percent (%) of the population lack access to basic public services of good quality. 4: Equal. Because of their social group, less than 5 percent (%) of the population lack access to basic public services of good quality.

v2svstterr — State authority over territory (loading +0.121)

Question. Over what percentage (%) of the territory does the state have effective control?

Responses. Percent.

v2dlunivl — Means-tested v. universalistic policy (loading +0.118)

Question. How many welfare programs are means-tested and how many benefit all (or virtually all) members of the polity?

Responses. 0: There are no, or extremely limited, welfare state policies (education, health, retirement, unemployment, poverty programs). 1: Almost all of the welfare state policies are means-tested. 2: Most welfare state policies means-tested, but a significant portion (e.g. 1/4 or 1/3) is universalistic and potentially benefits everyone in the population. 3: The welfare state policies are roughly evenly divided between means-tested and universalistic. 4: Most welfare state policies are universalistic, but a significant portion (e.g., 1/4 or 1/3) are means-tested. 5: Almost all welfare state policies are universal in character. Only a small portion is means-tested.

v2peapsgen — Access to public services distributed by gender (loading +0.118)

Question. Is access to basic public services, such as order and security, primary education, clean water, and healthcare, distributed equally according to gender?

Responses. 0: Extreme. Because of their gender, 75 percent (%) or more of women lack access to basic public services of good quality. 1: Unequal. Because of their gender, 25 percent (%) or more of women lack access to basic public services of good quality. 2: Somewhat Equal. Because of their gender, 10 to 25 percent (%) of women lack access to basic public services of good quality. 3: Relatively Equal. Because of their gender, 5 to 10 percent (%) of women lack access to basic public services of good quality. 4: Equal. Because of their gender, less than 5 percent (%) of women lack access to basic public services of good quality.

PC2 — top 15 negative loadings

These pull a country-year toward the negative pole of PC2. Several indicators here are coded so a *high* indicator score means *less* of the named feature — for example v2psparban (Party ban) runs 0 = “all parties banned” to 4 = “no parties banned”, so a high indicator score (no bans) contributes negatively to PC2, meaning *more* party bans pull

a country toward the *positive* pole of PC2. Each block below includes the full response scale so the directionality is unambiguous.

v2cademmob — Mobilization for democracy (loading -0.152)

Question. In this year, how frequent and large have events of mass mobilization for pro-democratic aims been?

Responses. 0: There have been virtually no events. 1: There have been several small-scale events. 2: There have been many small-scale events. 3: There have been several large-scale and small-scale events. 4: There have been many large-scale and small-scale events.

v2c1stown — State ownership of economy (loading -0.142)

Question. Does the state own or directly control important sectors of the economy?

Responses. 0: Virtually all valuable capital belongs to the state or is directly controlled by the state. Private property may be officially prohibited. 1: Most valuable capital either belongs to the state or is directly controlled by the state. 2: Many sectors of the economy either belong to the state or are directly controlled by the state, but others remain relatively free of direct state control. 3: Some valuable capital either belongs to the state or is directly controlled by the state, but most remains free of direct state control. 4: Very little valuable capital belongs to the state or is directly controlled by the state.

v2cavio1 — Political violence (loading -0.137)

Question. How often have non-state actors used political violence against persons this year?

Responses. 0: Not at all. Non-state actors did not use political violence. 1: Rare. Non-state actors rarely used political violence. 2: Occasionally. Non-state actors occasionally used political violence. 3: Frequently. Non-state actors frequently used political violence. 4: Often. Non-state actors often used political violence.

v2psparban — Party ban (loading -0.123)

Question. Are any parties banned?

Responses. 0: Yes. All parties except the state-sponsored party (and closely allied parties) are banned. 1: Yes. Elections are non-partisan or there are no officially recognized parties. 2: Yes. Many parties are banned. 3: Yes. But only a few parties are banned. 4: No. No parties are officially banned.

v2csantimv — CSO anti-system movements (loading -0.123)

Question. Among civil society organizations, are there anti-system opposition movements?

Responses. 0: No, or very minimal. Anti-system movements are practically nonexistent. 1: There is only a low-level of anti-system movement activity but it does not pose much of a threat to the regime. 2: There is a modest level of anti-system movement activity, posing some threat to the regime. 3: There is a high level of anti-system movement activity, posing substantial threat to the regime. 4: There is a very high level of anti-system movement activity, posing a real and present threat to the regime.

v2regoppgroupsize — Regime opposition groups size (loading -0.112)

Question. In total, how large is the share of the domestic adult (18+) population that are noteworthy opposition actors to the current political regime?

Responses. 0: Extremely small (About 1 percent of the population or less) 1: Very small (Between 1 percent and 5 percent of the population) 2: Small (Between 5 percent and 15 percent) 3: Moderate (Between 15 percent and 30 percent) 4: Large (More than 30 percent)

v2csprtcpt — CSO participatory environment (loading -0.112)

Question. Which of these best describes the involvement of people in civil society organizations (CSOs)?

Responses. 0: Most associations are state-sponsored, and although a large number of people may be active in them, their participation is not purely voluntary. 1: Voluntary CSOs exist but few people are active in them. 2: There are many diverse CSOs, but popular involvement is minimal. 3: There are many diverse CSOs and it is considered normal for people to be at least occasionally active in at least one of them.

v2mebias — Media bias (loading -0.110)

Question. Is there media bias against opposition parties or candidates?

Responses. 0: The print and broadcast media cover only the official party or candidates, or have no political coverage, or there are no opposition parties or candidates to cover. 1: The print and broadcast media cover more than just the official party or candidates but all the opposition parties or candidates receive only negative coverage. 2: The print and broadcast media cover some opposition parties or candidates more or less impartially, but they give only negative or no coverage to at least one newsworthy party or candidate. 3: The print and broadcast media cover opposition parties or candidates more or less impartially, but they give an exaggerated amount of coverage to the governing party or candidates. 4: The print and broadcast media cover all newsworthy parties and candidates more or less impartially and in proportion to their newsworthiness.

v2cacritic — Academics as critics (loading -0.109)

Question. To what extent do scholars and university students publicly criticize government policies?

Responses. 0: Not at all. Scholars and university students do not publicly express criticism of government policies. 1: To a small extent. Scholars and university students publicly express minor criticism of government policies. 2: To a moderate extent. Scholars and university students publicly express moderate criticism of government policies. 3: To a large extent. Scholars and university students publicly express substantive criticism of government policies. 4: To a major extent. Scholars and university students publicly express fundamental criticism of government policies.

v2mecrit — Print/broadcast media critical (loading -0.109)

Question. Of the major print and broadcast outlets, how many routinely criticize the government?

Responses. 0: None. 1: Only a few marginal outlets. 2: Some important outlets routinely criticize the government but there are other important outlets that never do. 3: All major media outlets criticize the government at least occasionally.

v2jureview — Judicial review (loading -0.104)

Question. Does any court in the judiciary have the legal authority to invalidate governmental policies (e.g. statutes, regulations, decrees, administrative actions) on the grounds that they violate a constitutional provision?

Responses. 0: No. 1: Yes.

v2psbars — Barriers to parties (loading -0.104)

Question. How restrictive are the barriers to forming a party?

Responses. 0: Parties are not allowed. 1: It is impossible, or virtually impossible, for parties not affiliated with the government to form (legally). 2: There are significant obstacles (e.g. party leaders face high levels of regular political harassment by authorities). 3: There are modest barriers (e.g. party leaders face occasional political harassment by authorities). 4: There are no substantial barriers.

v2meslfcen — Media self-censorship (loading -0.101)

Question. Is there self-censorship among journalists when reporting on issues that the government considers politically sensitive?

Responses. 0: Self-censorship is complete and thorough. 1: Self-censorship is common but incomplete. 2: There is self-censorship on a few highly sensitive political issues but not on moderately sensitive issues. 3: There is little

or no self-censorship among journalists.

v2cseeorgs — CSO entry and exit (loading -0.100)

Question. To what extent does the government achieve control over entry and exit by civil society organizations (CSOs) into public life?

Responses. 0: Monopolistic control. The government exercises an explicit monopoly over CSOs. The only organizations allowed to engage in political activity such as endorsing parties or politicians, sponsoring public issues forums, organizing rallies or demonstrations, engaging in strikes, or publicly commenting on public officials and policies are government-sponsored organizations. The government actively represses those who attempt to defy its monopoly on political activity. 1: Substantial control. The government licenses all CSOs and uses political criteria to bar organizations that are likely to oppose the government. There are at least some citizen-based organizations that play a limited role in politics independent of the government. The government actively represses those who attempt to flout its political criteria and bars them from any political activity. 2: Moderate control. Whether the government ban on independent CSOs is partial or full, some prohibited organizations manage to play an active political role. Despite its ban on organizations of this sort, the government does not or cannot repress them, due to either its weakness or political expedience. 3: Minimal control. Whether or not the government licenses CSOs, there exist constitutional provisions that allow the government to ban organizations or movements that have a history of anti-democratic action in the past (e.g. the banning of neo-fascist or communist organizations in the Federal Republic of Germany). Such banning takes place under strict rule of law and conditions of judicial independence. 4: Unconstrained. Whether or not the government licenses CSOs, the government does not impede their formation and operation unless they are engaged in activities to violently overthrow the government.

v2cainsaut — Institutional autonomy (loading -0.098)

Question. To what extent do universities exercise institutional autonomy in practice?

Responses. 0: No autonomy at all. Universities do not exercise any degree of institutional autonomy; non-academic actors control decision-making. 1: Minimal autonomy. Universities exercise only very limited institutional autonomy; non-academic actors interfere extensively with decision-making. 2: Moderate autonomy. Universities exercise some institutional autonomy; non-academic actors interfere moderately with decision-making. 3: Substantial autonomy. Universities exercise institutional autonomy to a large extent; non-academic actors have only rare and minimal influence on decision-making. 4: Complete autonomy. Universities exercise complete institutional autonomy from non-academic actors.

Loading mass by V-Dem category, per PC

The same redundancy worry as before: a PC that draws disproportionately from one V-Dem family is partly capturing module-internal covariance rather than generic institutional structure. The table below shows, for each of the first four PCs, the share of squared loading mass falling in each category.

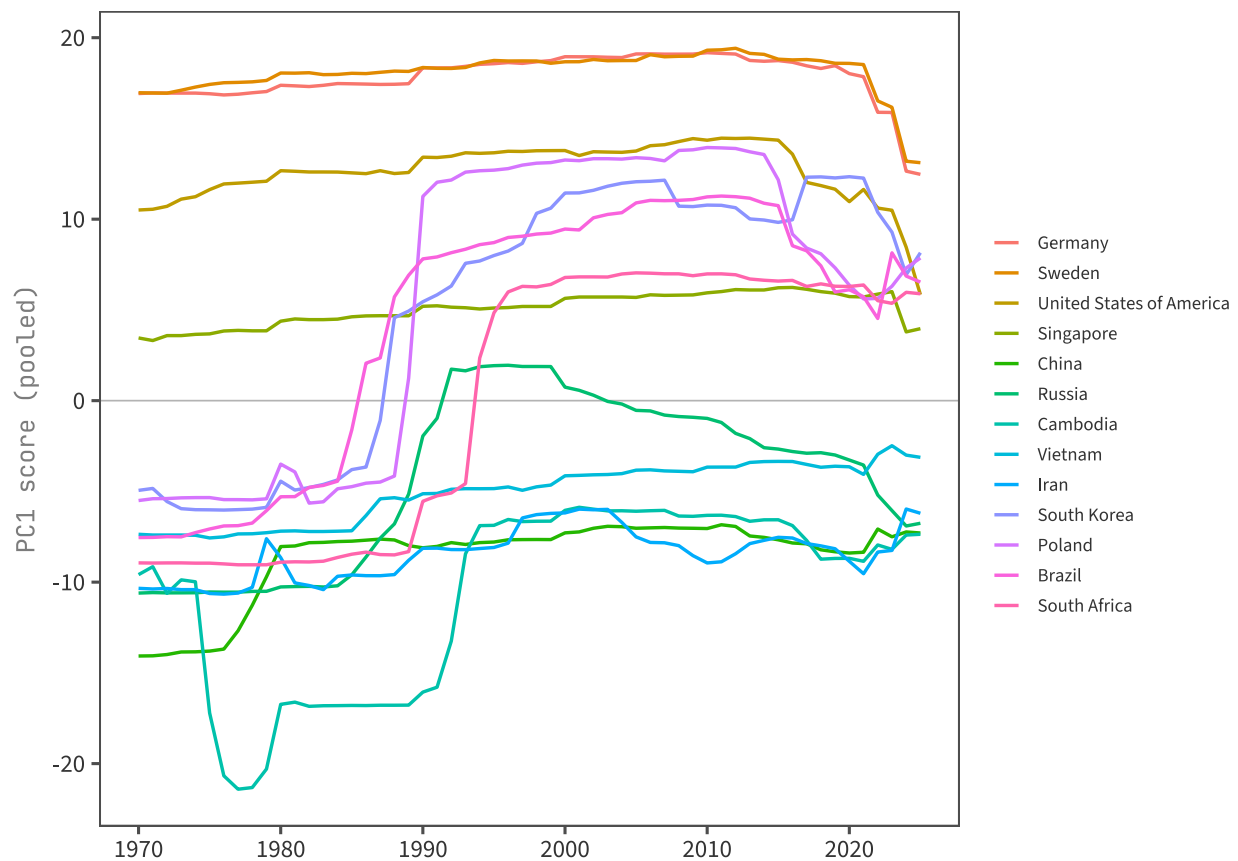
Table 3: Share of squared loading mass by V-Dem category, by PC.

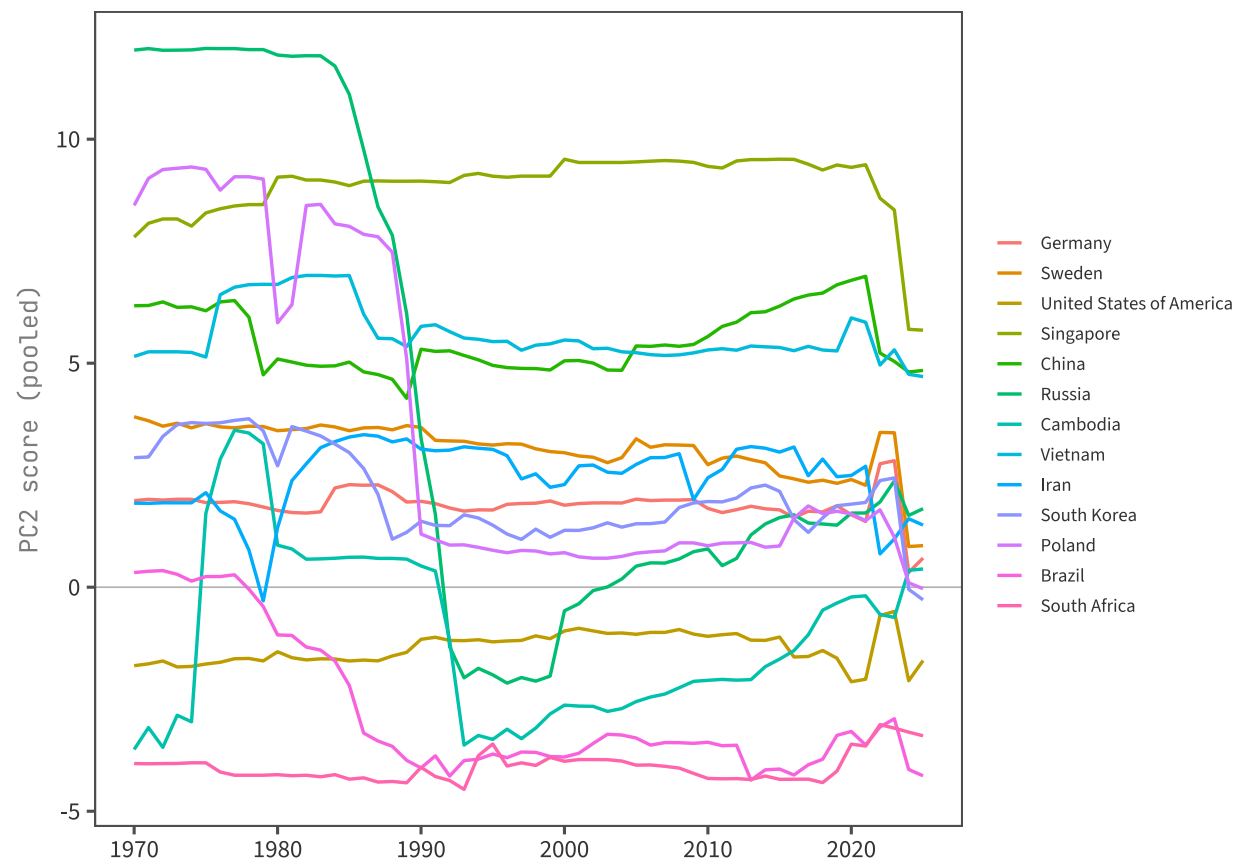
category	PC1	PC2	PC3	PC4
Media	9.5%	7.9%	5.5%	1.7%
Executive	8.5%	7.4%	19.5%	15.9%
Mobilization & Academic	7.5%	11.3%	9.5%	10.0%
Political Parties	5.5%	5.6%	8.5%	11.3%
Education	4.8%	4.3%	16.2%	6.8%
Judiciary	4.8%	4.6%	4.3%	3.9%

category	PC1	PC2	PC3	PC4
Legislature	4.7%	3.9%	3.3%	2.0%
Civil Society	4.5%	4.3%	1.9%	2.7%
Deliberation	4.2%	3.4%	1.8%	1.5%
State Capacity	2.2%	0.8%	0.8%	1.1%
Civil Liberties	15.5%	9.7%	6.2%	7.1%
Political Equality	14.9%	21.1%	8.1%	3.9%
Elections	11.8%	11.5%	9.8%	4.5%
Regime	0.9%	2.3%	2.4%	20.0%
Sovereignty	0.6%	1.7%	1.8%	7.6%
Technology & Media Environment	0.0%	0.2%	0.6%	0.2%

Sample country trajectories

A pooled PCA is only useful if its axes track the substantive history we think we know. We project a small panel of well-known countries into PC1 and PC2 over the full window.





Two-axis snapshot at year endpoints



Key takeaways: PC1 and PC2 as analytical tools

The pooled PCA produces dozens of components but only the first two carry enough variance and conceptual coherence to deploy on case studies. Both have a clear interpretation that survives careful reading of the loadings. Together they give a two-dimensional institutional space we can use on individual countries to separate regime-type movement from state-building movement. This section codifies what each axis means and how to read movement on it; downstream case-study writeups should refer back here rather than re-deriving the interpretation each time.

PC1: openness of the regime (the democracy axis)

Positive pole. Liberal democracy: freedom of expression and discussion, free press (low government censorship, low harassment of journalists, low media corruption), free academic and cultural expression, peaceful assembly, autonomous electoral management, no civil society repression, freedom from torture, transparent laws with predictable enforcement, access to justice, rigorous public administration, broad consultation in deliberation.

Negative pole. Personalist/ideological autocracy with a state-controlled information environment: state-owned print and broadcast media with political influence, control of entertainment content, political teacher firing/hiring with no teacher autonomy or independent unions, person-of-the-leader and ideological executive legitimation, HOS veto and dissolution power in practice, state-administered mass organizations, mobilization for autocracy.

How to read movement. A move on PC1 is regime change in the obvious sense — democratization (positive direction) or autocratization (negative direction). PC1 is the axis any V-Dem PCA recovers; the pooled fit just confirms it's robust across the 1970–2025 window and not an artifact of post-2000 measurement.

PC2: stateness — how much of social life the state encompasses

This is the axis whose interpretation took work to nail down (see the top-15 detail above). After fixing directionality on the reverse-coded indicators, PC2 contrasts two opposing **modes of governance** that cut across regime type.

Positive pole — “the state encompasses society.” Bundles two clusters that the data hold together:

1. *Broad, equal public-goods delivery.* Equal access to education, health, public services across income / urban-rural / social group / gender; state jobs accessible across income; universalistic rather than means-tested welfare. State authority over territory; uniform subnational administration of elections and civil liberties. Clean everyday administration (no legislative corruption, no vote-buying, no electoral violence).
2. *State control of the non-state sphere.* State-owned broadcast media; state ownership of the economy; biased state-aligned media; media self-censorship; government control over CSO entry and exit; low university autonomy; party bans and barriers to party formation; no judicial review constraining the executive.

These look like opposites at first — equality of delivery sounds liberal, control of the non-state sphere sounds illiberal — but the data bundle them because they share a common precondition: **infrastructural power** in Michael Mann's sense. A state with reach to deliver basic services uniformly across territory is the same state with reach to license every broadcaster, register every CSO, supervise every university, and gate every party-formation rule. Encompassment cuts both ways.

Negative pole — “society encompasses the state.” Private economy, private media, autonomous universities, autonomous CSOs, critical media, critical academics, judicial review constraining the executive, easy party formation and large opposition — alongside political violence by non-state actors, anti-system CSO movements, and mass mobilization for democracy. The unifying thread is *non-state actors having visible weight*, whether that weight

takes benign-constitutional form (a constitutional court overriding the executive) or violent-contentious form (insurgent movements attacking it). PC2 lumps them because both register as the state's encompassment of society being incomplete.

How to read movement.

- Positive PC2 movement = state-building / state-encompassment deepening (administrative reach into delivery + control of the non-state sphere).
- Negative PC2 movement = state-encompassment shrinking, either through deliberate liberal-pluralist constraint (judicial review, free CSOs) or through state weakening (non-state violence, contested authority).

PC2 is **orthogonal to PC1 by construction**. Welfare democracies and developmental autocracies sit together at high PC2 (they share encompassment despite opposing each other on PC1). Liberal-market democracies and weak/contested polities sit together at low PC2 (they share non-state weight on the state, despite opposing each other on PC1). This is the empirical signature of the long-standing political-theory claim (Tilly, Mann, Migdal) that state capacity is a separate dimension from regime type.

How to deploy PC1 and PC2 in case studies

When projecting a country into this space, report and read both axes separately. The diagnostic patterns to look for:

- **Regime change with stable stateness.** PC1 moves, PC2 flat. Many third-wave democratization cases — institutions liberalize without the state's reach into economy, media, civil society shifting much.
- **State-building without democratization.** PC2 up, PC1 flat. The developmental-state pattern — encompassment deepens (administrative delivery + control of non-state sphere) without regime opening. Singapore, post-Doi Moi Vietnam, late-Hun-Sen Cambodia are candidates.
- **State collapse with regime liberalization.** PC2 down, PC1 up. The post-Soviet 1990s pattern — media, parties, civil society, and economy de-state-ize while the regime formally opens. Russia 1989–1996 is the textbook case.
- **State reconsolidation under autocracy.** PC2 up, PC1 down. The recovery from the previous pattern — regime closes, state reabsorbs media, parties, CSOs. Russia 2000s.

A case-study writeup should locate the country in (PC1, PC2) space at endpoints, plot the trajectory, and decompose any large move into its PC1 and PC2 components. The four patterns above are the ones to look for; their absence is just as informative as their presence.

Persist outputs

```
## Wrote: outputs/pooled_pca/pca_pool.rds
```

```
## Wrote: outputs/pooled_pca/country_year_pc_scores.csv (9528 rows)
```

```
## Wrote: outputs/pooled_pca/pca_loadings_pc1_pc4.csv
```

Notes and what to read into this

- **Sign convention.** PC sign is arbitrary in prcomp. We don't flip anything in this file; if a downstream caller wants "high PC1 = liberal democracy", inspect the top loadings of PC1 above and apply a sign flip when read-

ing the saved scores. The reproducible move is to store both the rotation and the sign convention you settled on.

- **Pooled vs. snapshot PCAs are answering different questions.** Pooled = “what are the dominant institutional axes across our development-relevant window?” Snapshot = “what’s the shape of institutional space *now*?” The 2025 snapshot embeds modules (Digital Society) that are absent from most of the pool. The trajectory chart in `case-studies/cambodia/cambodia-analysis.Rmd` should switch to the pooled rotation if its claim is “Cambodia’s institutions over time.”
- **Coverage filter.** The 50% pool-coverage threshold is the lever that decides whether `v2sm*` enters the PCA. With 50% almost all `v2sm` indicators get dropped (they’re observed in ~46% of cells). Tightening to 70% is conservative; loosening to 30% would re-admit them. We start at 50% and report the count above; the file is set up so the threshold is a single-line change.
- **Imputation.** Pool-mean imputation makes the `v2sm` pre-2000 cells approximately constant at the pool mean — neutral by construction but it’s still synthetic data. If we wanted to be stricter we’d drop any indicator whose post-2000 mean differs materially from its full-pool mean (which would catch indicators where pre-2000 cells have been imputed to a value far from where the indicator actually sits when measured). Worth doing as a sensitivity if PC2’s interpretation hinges on a specific `v2sm` indicator that the filter let through.